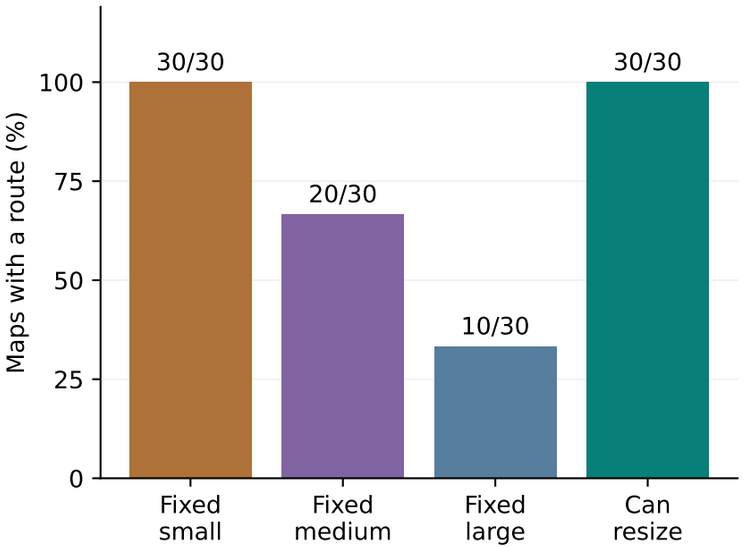
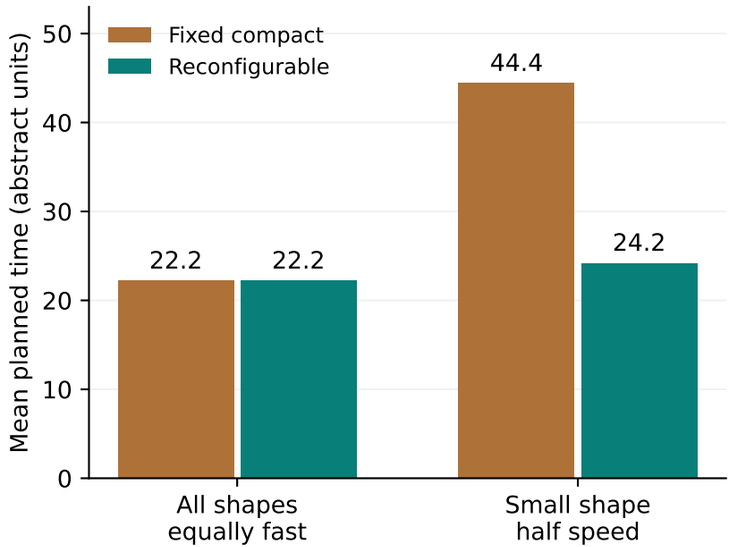
When does changing shape help?

Geometric navigation baseline | Walls with passages and no detour

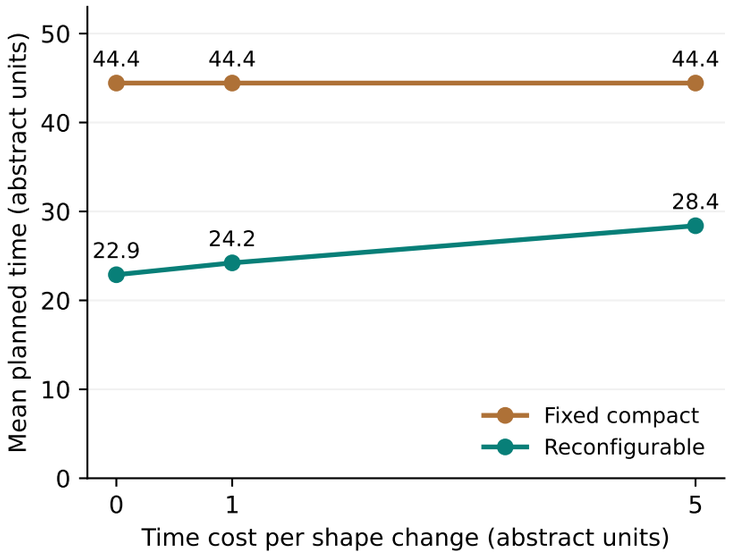
1. Fitting through the openings



2. A speed tradeoff creates value



3. Slower reshaping reduces the gain



Panels 2–3 compare the same 30 maps; both robots succeed on all 30. Panel 3 assumes half-speed small bodies.

Assumed speeds, full-map planning, and free initial shape choice. No physics, learning, or hardware validation.

Deterministic designed cases; no statistical error bars. Medium/large travel times omitted because their successful map sets differ.